

Exercise 1 Let $[x]$ denote the integer part of $x \in \mathbb{R}$, and let $\alpha > 0$.

a) Prove that $f_\alpha: [1, \infty) \rightarrow \mathbb{R}^+$ defined by

$$f_\alpha(x) = \left(\frac{1}{[x]}\right)^\alpha$$

is $B(\mathbb{R})$ -measurable

b) Calculate $\int_{[1, \infty)} f_\alpha d\lambda$, i.e. write it as an infinite sum.

(Hint; Use MCT.)

c) For which α is $\int_{[1, \infty)} f_\alpha d\lambda < \infty$?

(You may use that $\underbrace{\int_{[a, b]} f d\lambda}_{\text{Lebesgue}} = \underbrace{\int_a^b f(x) dx}_{\text{Riemann}}$

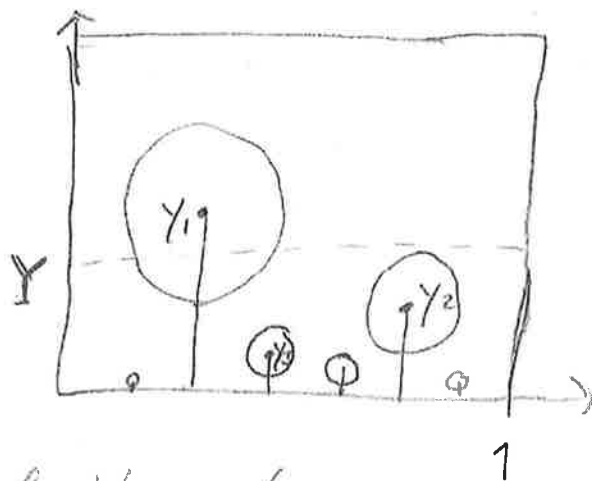
for continuous f .)

Exercise 2 Consider the square $[0,1]^2$

This is a cheese with holes.

Each hole sits at height y_k with radius $y_k/2$.

Suppose that $\sum_{k=1}^{\infty} y_k < \infty$



Consider the lower part of the cheese

$[0,1] \times [0,Y]$, where $0 < Y < 1$. Prove that the density d_Y of the cheese in this part approaches 1 as $Y \rightarrow 0$. Hints;

a) Prove $d_Y = \frac{\text{Area of cheese in } [0,1] \times [0,Y]}{1 \cdot Y} \geq 1 - \frac{\frac{\pi}{4} \sum_{y_k \leq 2Y} y_k^2}{Y}$

b) Conclude that you need to show

$$\lim_{Y \rightarrow 0} \sum_{y_k \leq Y} y_k^2 / Y = 0 \quad (1)$$

c) Let μ be the measure on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$

defined by $\mu(\{j\}) = y_j$, let Y_n be s.t.

$$\lim_{n \rightarrow \infty} Y_n = 0$$

and set $f_n: \mathbb{N} \rightarrow \mathbb{R}^+$

via $f_n(j) = \begin{cases} y_j / Y_n & y_j \leq Y_n \\ 0 & \text{else} \end{cases}$. Apply DCT to get (1).

Exercise 2' Given $\alpha < 2$ find $y_n > 0$
s.t. $\sum_{k=1}^{\infty} y_k < \infty$ but

$$\lim_{\gamma \rightarrow 0} \frac{\sum_{y_n < \gamma} y_n^{\alpha}}{\gamma} = \infty$$

Exercise 3 Extend DCT as follows;
If $\varphi_{\alpha} \in L^1(X, \mathcal{A}, \mu) \quad \forall \alpha \in [0, 1)$ is such
that $\alpha \mapsto \varphi_{\alpha}(x)$ is continuous $\forall x \in X$,
and $\exists g \in L^1(X, \mathcal{A}, \mu)$ with
 $|\varphi_{\alpha}(x)| \leq g(x) \quad \forall \alpha \quad \forall x,$

then

$$\lim_{\alpha \rightarrow 0} \int \varphi_{\alpha} d\mu = \int \varphi_0 d\mu$$

Hint Argue by contradiction to extract
a sequence that contradicts DCT.

Exercise 4. Calculate

$$\int \frac{t}{1+(xt)^2} d\lambda(x) \quad (t > 0)$$

Calculate $\lim_{t \rightarrow 0} \frac{t}{1+(xt)^2}$

With suitable choice of $(X, \mathcal{A}, \mu)$ & φ_2 , show that $|\varphi_2(x)| \leq g(x)$ can not be omitted from Exercise 3.

Exercise 5 Let $f: X \times \mathbb{R} \rightarrow \mathbb{R}$ be such that $\partial_x f(t, x)$ exists $\forall t \in X$. Assume

$$\exists g \in L^1(X, \mathcal{A}, \mu) \text{ s.t. } |\partial_x f(t, x)| \leq g(t)$$

$$\forall 0 < x < 1. \text{ Set } F(x) = \int f(t, x) d\mu(t)$$

and assume that $F(x)$ is well-defined (ie $f(\cdot, x) \in L^1(X, \mathcal{A}, \mu) \forall x$). Prove that

$F'(x)$ exists $\forall 0 < x < 1$ and that

$$F'(x) = \int \partial_x f(t, x) d\mu(t)$$

Exercise 6. Let $\gamma: [0,1] \rightarrow \mathbb{C}$ be injective, $\gamma(0) = \gamma(1)$ & continuously differentiable. Set $\Gamma = \gamma([0,1])$. Given a continuous function φ on Γ , we define $h: \mathbb{C} \setminus \Gamma \rightarrow \mathbb{C}$ by

$$h(z) = \int_{\Gamma} \frac{\varphi(\zeta)}{\zeta - z} \frac{d\zeta}{2\pi i} = \int_{[0,1]} \frac{\varphi(\gamma(t))}{\gamma(t) - z} \frac{\gamma'(t)}{2\pi i} dt$$

Write $z = x + iy$. Show that

$$\partial_x h(z) = \int_{\Gamma} \frac{\varphi(\zeta)}{(\zeta - z)^2} \frac{d\zeta}{2\pi i}, \quad \forall z \notin \Gamma.$$

Exercise 7. A function is called

analytic if it satisfies

Cauchy-Riemann's equation

$$\partial_x h = -i \partial_y h$$

Is h analytic?

Answers 1

1a) $f_\alpha^{-1}([t, \infty))$ is a half-axis

1b) Set $f_k = f_\alpha \chi_{[1, k]}$. Then

$$\int f_\alpha d\lambda = \lim_{k \rightarrow \infty} \int f_k d\lambda = \lim_{k \rightarrow \infty} \sum_{j=1}^{k-1} \frac{1}{j^\alpha} = \sum_{j=1}^{\infty} \frac{1}{j^\alpha}$$

c)

I just wanted you to compare

$$\sum_{j=1}^{\infty} \frac{1}{j^\alpha} \text{ with } \int_1^{\infty} \frac{1}{x^\alpha} dx, \text{ as in basic}$$

analysis. Answer $\alpha > 1$.

2a) $\frac{\pi}{4} \sum_{\gamma_k \leq Y} \gamma_k^2$ is the area of all holes with a piece below Y ...

2c) We have $|f_k| \leq 1$ and $\int 1 d\mu < \infty$.

$\gamma_k \leq Y$ We get by MCT

$$\sum_{\gamma_j \leq Y_k} \gamma_j^2 = \int f_k d\mu \text{ and by}$$

$$\text{DCT we have } \lim_{k \rightarrow \infty} \int f_k d\mu = \int \lim_{k \rightarrow \infty} f_k d\mu = \int 0 d\mu = 0.$$

2) Suppose $\alpha > 1$.
 Pick $\beta > 1$ and $y_k = \frac{1}{k^\beta}$. We need to
 show $\lim_{k \rightarrow \infty} \frac{\sum_{j=k}^{\infty} y_j^\alpha}{y_k} = \lim_{k \rightarrow \infty} \sum_{j=k}^{\infty} \frac{k^\beta}{j^{\alpha\beta}} = \infty$

As in Exercise 1 this is true if

$$\lim_{k \rightarrow \infty} \int_k^{\infty} \frac{k^\beta}{x^{\alpha\beta}} dx = \lim_{k \rightarrow \infty} \left[\frac{k^\beta x^{1-\alpha\beta}}{1-\alpha\beta} \right]_k^{\infty} = \infty$$

If $\alpha\beta > 1$ the bracket evaluates to
 $\frac{k^{\beta+1-\alpha\beta}}{\alpha\beta-1}$ and we want to have

$\beta+1-\alpha\beta < 0 \Leftrightarrow \beta < \frac{1}{\alpha-1}$. Since $\alpha < 2$
 such a $\beta > 1$ can be chosen.

Finally, if $\alpha < 1$ we can pick $\tilde{\alpha} > 1$ and
 y_k as above. Then

$$\lim_{k \rightarrow \infty} \frac{\sum_{j=k}^{\infty} y_j^\alpha}{y_k} \geq \lim_{k \rightarrow \infty} \frac{\sum_{j=k}^{\infty} y_j^{\tilde{\alpha}}}{y_k} = \infty$$

3) Do as the hint says.

$$4 \quad \int \frac{t}{1+(xt)^2} d\mu(x) = \left[\arctan(tx) \right]_{-\infty}^{\infty} = \pi$$

and $\lim_{t \rightarrow 0} \frac{t}{1+(xt)^2} = 0$ so with

$$X = \mathbb{R}, \mathcal{A} = \mathcal{B}(\mathbb{R}), \mu = dx \quad \& \quad \varphi_\alpha(x) = \frac{\alpha}{1+(2x)^2}$$

We have $\lim_{\alpha \rightarrow 0} \int \varphi_\alpha d\mu = \pi$ but $\int \varphi_0 d\mu = 0$

5 Im sorry, the role of x & t have been swapped in this answer...

$$F(t+h) - F(t) = \int f(x, t+h) - f(x, t) d\mu(x)$$

$$= \int \partial_t f(x, t + \theta_{x,h} h) dh d\mu(x)$$

where $0 < \theta_{x,h} < 1$, (by some consequence of the mean value theorem). Setting

$\varphi_h = (f(x, t+h) - f(x, t))/h$ we thus have

$$|\varphi_h(x)| \leq \partial_t f(x, t + \theta_{x,h} h) \leq g(x) \quad \text{and}$$

$$\lim_{h \rightarrow 0} \frac{F(t+h) - F(t)}{h} = \lim_{h \rightarrow 0} \int \varphi_h d\mu = \int \lim_{h \rightarrow 0} \varphi_h(x) d\mu$$

$$= \int \partial_t f(x, t) d\mu(x) \quad \text{by Exercise 4.}$$

6 Of course, differentiation of complex-valued functions is done as in real analysis by working with the real and imaginary part separately. First show that $\partial_x \left(\frac{\alpha}{\beta-x} \right) = \frac{\alpha}{(\beta-x)^2}$ for any $\alpha, \beta \in \mathbb{C}$.

(Hint, the expansion $\frac{1}{1-z} = \sum_{n=1}^{\infty} z^n$, valid for all $z \in \mathbb{C}$ with $|z| < 1$, provides a shortcut)

Now set $f(t, x) = \frac{\varphi(\gamma(t))}{\gamma(t) - (x+iy)} \frac{\gamma'(t)}{2\pi i}$ and show that

$$a) \partial_x f(t, x) = \frac{\varphi(\gamma(t))}{(\gamma(t) - (x+iy))^2} \frac{\gamma'(t)}{2\pi i}$$

b) Exc 5 holds with the obvious modifications for complex-valued functions and any interval I in place of $(0,1)$.

Finally fix a point $z_0 = x_0 + iy_0 \notin \Gamma$ where we want to differentiate, let ε be the radius of a ball with center z_0 that does not intersect Γ , and set $I = (x_0 - \varepsilon, x_0 + \varepsilon)$.



Show that there must exist a constant B such that

$$\left| \frac{\varphi(\gamma(t))}{\gamma(t) - (x + iy_0)} \right| \leq B$$

for all $t \in [0, 1]$ and all $x \in I$.

Finally apply (the modified version of)

Exc 5 with $g(t) = \frac{B}{2\pi} |\gamma'(t)|$ and $F = h$.

7 Yes, upon modification of the argument in 6 we can also show that

$$\partial_{\bar{y}} h(z_0) = i \int_{\Gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^2} \frac{d\zeta}{2\pi i}$$

